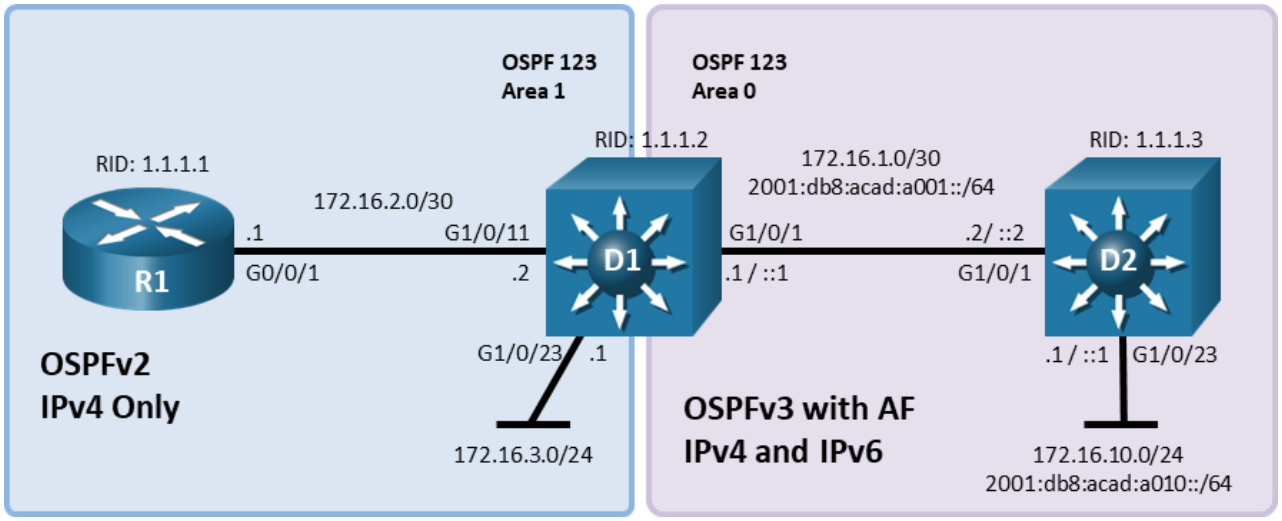
Lab - Configure OSPF Authentication

Topology



Addressing Table

Device	Interface	IPv4 Address	IPv6 Address	IPv6 Link-Local
R1	G0/0/1	172.16.2.1/30	N/A	N/A
D1	G1/0/1	172.16.1.1/30	2001:db8:acad:a001::1/64	fe80::d1:1
	G1/0/11	172.16.2.2/30	N/A	N/A
	G1/0/23	172.16.3.1/24	N/A	N/A
D2	G1/0/1	172.16.1.2/30	2001:db8:acad:a001::2/64	fe80::d2:1
	G1/0/23	172.16.10.1/24	2001:db8:acad:a010::1/64	fe80::d2:2

Objectives

Part 1: Build the Topology and Configure Basic Device Settings and IP Addressing

Part 2: Different Authentication Types for OSPFv2 and OSPFv3

Part 3: OSPFv2 Type 1 Clear Text Authentication

Part 4: OSPFv2 Type 2 MD5 Authentication

Part 5: OSPFv3 IPsec AH (Authentication & Integrity)

Part 6: OSPFv3 IPsec ESP (encryption + authentication)

Part 7: OSPFv3 Authentication Trailer (RFC 7166)

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Background / Scenario

In this topology, all devices have been configured with the appropriate interface addressing. **R1** and **D1** are running **OSPFv2 for IPv4**, forming Area 1 and providing IPv4-only routing for that portion of the network. **D1** and **D2** are running **OSPFv3 with address families**, supporting both **IPv4** and **IPv6 routing**.

In this lab, Area 1 is intentionally configured using **OSPFv2 for IPv4 only** to demonstrate OSPFv2 authentication. As a result, IPv4 routes from Area 1 are isolated from the rest of the routing domain. Redistribution between OSPFv2 and OSPFv3 is possible on some platforms, but it adds complexity and risks (loops/route feedback), so it is intentionally avoided here.

D1 and **D2** (Cisco 9300 multilayer layer switches) are used to demonstrate OSPFv3 IPsec-based authentication because the router platforms, **R1** (Cisco 8200/8300) and IOS-XE image used in this topology **do not support IPsec AH or ESP configurations for OSPFv3**. This is not uncommon and is explained in **Part 7: OSPFv3 Authentication Traile**r. Both D1 and D2 support all required OSPFv3 authentication mechanisms, allowing the lab to demonstrate IPsec AH, IPsec ESP, and the Authentication Trailer consistently.

Note: The routers used with this CCNP lab is a Cisco 8200L and the two Layer 3 switches are Catalyst 9300L switches using Cisco IOS XE Release 17.12.05 (universalk9 image). Other routers and Layer 3 switches and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Ensure that the routers and switches have been erased and have no startup configurations. If you are unsure contact your instructor.

Required Resources

- 1 Router (Cisco Catalyst 8200L with Cisco IOS XE Release 17.12.05c universal image or comparable)
- 2 Switches (Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.05 universal image or comparable)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

In Part 1, you will set up the network topology and configure basic settings and interface addressing on the router and switches.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram and cable as necessary.

Step 2: Configure basic settings for each router.

a. Console into each device, enter global configuration mode, and apply the basic settings and interface addressing using the following startup configurations for each device.

Router R1

```
hostname R1
!
no ip domain lookup
line con 0
```

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```
exec-timeout 0 0
logging sync
!
interface GigabitEthernet0/0/1
 ip address 172.16.2.1 255.255.255.252
 ip ospf 123 area 1
 ip ospf network point-to-point
 no shutdown
 exit
!
router ospf 123
 router-id 1.1.1.1
 exit
!
end
```

Switch D1

```
hostname D1
!
no ip domain lookup
line con 0
exec-timeout 0 0
logging sync
!
ip routing
ipv6 unicast-routing
!
interface GigabitEthernet1/0/1
 no switchport
 ip address 172.16.1.1 255.255.255.252
 ipv6 address 2001:db8:acad:a001::1/64
 ipv6 address fe80::d1:1 link-local
 ospfv3 123 ipv6 area 0
 ospfv3 123 ipv4 area 0
 ospfv3 network point-to-point
 no shutdown
 exit
!
interface GigabitEthernet1/0/11
 no switchport
 ip address 172.16.2.2 255.255.255.252
 ip ospf 123 area 1
 ip ospf network point-to-point
 no shutdown
 exit
```

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```
!
interface GigabitEthernet1/0/23
 no switchport
 ip address 172.16.3.1 255.255.255.0
 ip ospf 123 area 1
 no shutdown
 exit
!
router ospf 123
 router-id 1.1.1.2
 exit
!
router ospfv3 123
 !
 address-family ipv4 unicast
  router-id 1.1.1.2
 exit-address-family
 !
 address-family ipv6 unicast
  router-id 1.1.1.2
 exit-address-family
!
end
```

Switch D2

```
hostname D2
!
no ip domain lookup
line con 0
exec-timeout 0 0
logging sync
!
ip routing
ipv6 unicast-routing
!
interface GigabitEthernet1/0/1
 no switchport
 ip address 172.16.1.2 255.255.255.252
 ipv6 address 2001:db8:acad:a001::2/64
 ipv6 address fe80::d2:1 link-local
 ospfv3 123 ipv6 area 0
 ospfv3 123 ipv4 area 0
 ospfv3 network point-to-point
 no shutdown
 exit
!
```

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```
interface GigabitEthernet1/0/23
 no switchport
 ip address 172.16.10.1 255.255.255.0
 ipv6 address 2001:db8:acad:a010::1/64
 ipv6 address fe80::d2:2 link-local
 ospfv3 123 ipv6 area 0
 ospfv3 123 ipv4 area 0
 no shutdown
 exit
!
router ospfv3 123
 !
 address-family ipv4 unicast
  router-id 1.1.1.3
 exit-address-family
 !
 address-family ipv6 unicast
  router-id 1.1.1.3
 exit-address-family
!
end
```

b. Save the running configuration to startup-config.

Part 2: Different Authentication Types for OSPFv2 and OSPFv3

The table below provides a **high-level comparison** of the authentication mechanisms available in OSPFv2 and OSPFv3. At this point, you are **not expected to fully understand every option** listed. Instead, this table serves as a roadmap that shows how authentication has evolved from OSPFv2 to OSPFv3.

As you progress through the lab, you will configure and verify **each authentication method step by step**. After completing those sections, this table should make much more sense and help reinforce where each method fits, what it protects, and how OSPFv3 differs architecturally from OSPFv2—particularly in how authentication is implemented outside the OSPF packet header.

Keep this table in mind as a reference and revisit it after completing each authentication lab to solidify your understanding.

Feature	OSPFv2	OSPFv3
Authentication location	Built into OSPF header	External to OSPF header (IPsec or Authentication Trailer)
Authentication types	**Type 0** (None) **Type 1** (Simple cleartext password) **Type 2** (Cryptographic MD5)	**None** (no authentication configured) **IPsec AH** (authentication & integrity) **IPsec ESP** (encryption + authentication) **Authentication Trailer** (RFC 7166)
OSPF packets authenticated	All OSPF control packets	All OSPF control packets

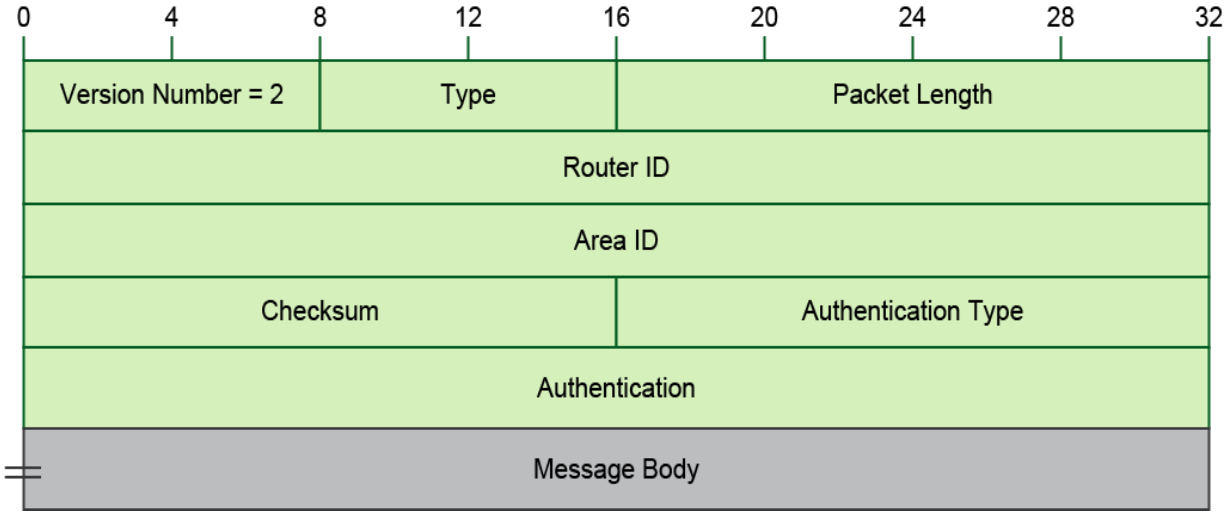
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Regardless of the authentication method used, OSPFv2 and OSPFv3 authenticate *all OSPF control packets* (OSPF Hello, DBD, LSR, LSU, and LSA packets), ensuring the integrity of the entire routing exchange.

The following table summarizes the key differences between OSPFv2 and OSPFv3 authentication mechanisms. It is intended as a reference overview and will become clearer as you configure and verify each authentication method throughout the lab.

Authentication Method	OSPFv2	OSPFv3
Cleartext password	Yes (**Type 1**)	No
Cryptographic options	MD5 only (**Type 2**)	HMAC-SHA-1 / SHA-256 / SHA-384 / SHA-512
Encryption support	No	Yes (IPsec ESP)
IPv4 support	Yes	Yes (via address families)
IPv6 support	No	Yes (native)

OSPFv2 embeds authentication directly in OSPFv2 packet header, but its security options are limited and outdated.



OSPFv3 intentionally removed authentication from the protocol itself, instead relying on modern security mechanisms such as IPsec and SHA-based cryptographic authentication. This lab explores those mechanisms and shows how OSPFv3 provides significantly stronger and more flexible protection for routing infrastructure.

When IPsec is used with OSPFv3, the entire IPv6 packet, including the OSPFv3 header and payload, is protected by IPsec. Authentication and/or encryption are applied outside the OSPFv3 protocol itself, using IPv6 extension headers rather than fields within the OSPF packet.

Note: OSPFv3 and IPv4-Only Areas

When OSPFv3 is used to advertise **IPv4 routes**, the protocol itself **still runs over IPv6**. This is true even in an IPv4-only OSPFv3 area or "island." In such a design, IPv6 is used solely as the transport for OSPFv3 control traffic. IPv6 routing does not need to be enabled for end-to-end forwarding.

To deploy **OSPFv3 for IPv4 routing** in an **IPv4-only area** (for example if we were to use OSPFv3 in Area 1), the following minimum requirements must be met:

- **IPv6 unicast routing** must be enabled.

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- **IPv6 link-local addresses** must exist on participating interfaces (automatically or manually configured).
- **OSPFv3** must be configured with the **IPv4 address family**.

Under these conditions, OSPFv3 can successfully form neighbor adjacencies and exchange IPv4 routing information, even though IPv6 is not used for data traffic.

Part 3: OSPFv2 Type 1 Clear Text Authentication

By default, OSPF does not use authentication. In this first part of the lab, you will configure **OSPFv2 simple password authentication**, also known as **cleartext or Type 1 authentication**, on routers **R1** and **D1**. This authentication method protects OSPF neighbor relationships by requiring participating routers on a link to share a common password before exchanging routing information.

With **Type 1** authentication, the password is **carried in cleartext within the OSPF packet header**. Because the authentication key is not encrypted, this method is vulnerable to packet sniffing and should not be used in production networks. However, it remains useful for learning purposes and for understanding how OSPF authentication operates at a fundamental level.

To enable OSPFv2 cleartext authentication, two steps are required:

1. **Configure the authentication key (password)** on the interface using the **ip ospf authentication-key** command.
2. **Enable authentication**, either on the interface itself or for the entire OSPF area.

The password is inserted directly into the OSPF header when routing protocol packets are sent. All routers on the same network segment must be configured with the **exact same password** for OSPF neighbor adjacencies to form. Passwords are limited to **eight characters**, and only the first eight characters are used if a longer string is entered.

The password that is created by this command is used as a "key" that is inserted directly into the OSPF header when Cisco IOS Software originates routing protocol packets. A separate password can be assigned to each network on a per-interface basis. All neighboring routers on the same network must have the same password to be able to exchange OSPF information.

Step 1: Verify OSPFv2 neighbor adjacency between R1 and D1.

Before configuring authentication, confirm that R1 and D1 have successfully formed an **OSPFv2 neighbor relationship** and are in the FULL state.

```
R1# show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.2           0   FULL/  -        00:00:37    172.16.2.2      GigabitEthernet0/0/1

D1# show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:30    172.16.2.1      GigabitEthernet1/0/11
```

Step 2: On R1, set the key and enable authentication on the link.

Configure the shared OSPF authentication key and enable OSPFv2 authentication on the interface connected to D1.

```
R1(config)# interface g0/0/1
R1(config-if)# ip ospf authentication-key PLAINTXT
R1(config-if)# ip ospf authentication
```

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OSPFv2 authentication can be enabled **either on a per-interface basis or for an entire OSPF area**. When authentication is configured at the interface level, it applies only to that specific link. When configured at the area level, authentication is required on all OSPF-enabled interfaces within the area, unless explicitly overridden.

Note: If the OSPF Dead Timer expires before you configure authentication on D1, there will be a loss of adjacency, and you will see the following message:

```
*Jan 19 16:09:00.944: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.2 on GigabitEthernet0/0/1
from FULL to DOWN, Neighbor Down: Dead timer expired
```

Step 3: On D1, set the key on the link and enable authentication in OSPFv2 for Area 1.

Configure the **OSPFv2 authentication key** on the interface connecting D1 to R1 and then enable **OSPF authentication for Area 1**. Applying authentication at the area level ensures that all OSPF-enabled interfaces in Area 1 require authentication, unless explicitly overridden on a specific interface.

```
D1(config)# interface g1/0/11
D1(config-if)# ip ospf authentication-key PLAINTXT
D1(config-if)# exit
D1(config)# router ospf 123
D1(config-router)# area 1 authentication
D1(config-router)# end
```

Note: If configured correctly, you will see the following message as adjacency is reestablished:

```
*Jan 19 16:05:59.626: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.1 on
GigabitEthernet1/0/11 from LOADING to FULL, Loading Done
```

Step 4: Verify OSPFv2 neighbor adjacency between R1 and D1 after authentication.

After configuring OSPFv2 authentication, verify that the neighbor adjacency between R1 and D1 has successfully re-established.

```
R1# show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.2           0   FULL/  -        00:00:35    172.16.2.2      GigabitEthernet0/0/1

D1# show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:31    172.16.2.1      GigabitEthernet1/0/11
```

Step 5: Verify OSPFv2-learned route in R1's IPv4 routing table.

With the OSPFv2 neighbor adjacency established, verify that R1 has learned the **remote IPv4 network from D1** and that it appears in R1's IPv4 routing table as an OSPF route.

```
R1# show ip route ospf | begin Gateway
Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 6 subnets, 3 masks
O        172.16.3.0/24 [110/2] via 172.16.2.2, 00:05:19, GigabitEthernet0/0/1
```

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Step 6: Enable service password-encryption.

a. If the **service password-encryption** command is not used when configuring OSPF authentication, the key will be stored as plaintext in the router configuration.

```
R1# show running-config | section interface GigabitEthernet0/0/1
interface GigabitEthernet0/0/1
 ip address 172.16.2.1 255.255.255.252
 ip ospf authentication
 ip ospf authentication-key PLAINTXT
 ip ospf network point-to-point
 ip ospf 123 area 1
 negotiation auto
```

b. If you configure the **service password-encryption** command, the key will be stored and displayed in an encrypted form. When it is displayed, there will be an encryption type of 7 that is specified before the encrypted key. However, this encryption is easy to crack. Search for “type 7 password decrypt” to find a site where you can verify how quickly type 7 is cracked.

```
R1# conf t
R1(config)# service password-encryption
R1(config)# end
R1# show running-config | section interface GigabitEthernet0/0/1
interface GigabitEthernet0/0/1
 ip address 172.16.2.1 255.255.255.252
 ip ospf authentication
 ip ospf authentication-key 7 053B2A2E08627A312D
 ip ospf network point-to-point
 ip ospf 123 area 1
 negotiation auto
```

Step 7: Verify OSPFv2 Type 1 authentication on the interface.

Use **show ip ospf interface** to verify that OSPFv2 Type 1 (simple password) authentication is enabled on the interface. The line **Simple password authentication enabled** confirms that cleartext authentication is active on this interface.

```
D1# show ip ospf interface g 1/0/11
GigabitEthernet1/0/11 is up, line protocol is up (connected)
  Internet Address 172.16.2.2/30, Interface ID 40, Area 1
  Attached via Interface Enable
  Process ID 123, Router ID 1.1.1.2, Network Type POINT_TO_POINT, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Enabled by interface config, including secondary ip addresses
  Transmit Delay is 1 sec, State POINT_TO_POINT
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  <output omitted>
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.1
  Suppress hello for 0 neighbor(s)
  Simple password authentication enabled
```

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Security Note: OSPFv2 Type 1 (simple password) authentication sends the authentication key **in cleartext within the OSPF packet header**. This makes it vulnerable to packet capture and replay attacks, as any device with access to the link can read the password. For this reason, Type 1 authentication should be used **only for learning or legacy compatibility** and is not recommended for production networks.

Part 4: OSPFv2 Type 2 MD5 Authentication

In this part, you will configure **OSPFv2 Type 2 authentication**, which uses **MD5 cryptographic hashing** to authenticate OSPF packets exchanged between neighboring routers. Unlike Type 1 (simple password) authentication, the shared key is **not sent in cleartext**. Instead, each OSPF packet includes a cryptographic hash that is generated using the packet contents and the shared secret key.

With MD5 authentication, routers validate the integrity and authenticity of OSPF packets by **independently computing the hash and comparing it to the value received from the neighbor**. This protects OSPF routing exchanges from passive packet sniffing and simple spoofing attacks, making it significantly more secure than cleartext authentication.

Step-by-Step (OSPFv2 Type 2 MD5)

The router takes the OSPF packet contents and…

1) **Combines** them with the **shared secret key**
2) Runs the **MD5 algorithm**
3) Produces a fixed-length output, which is the **message digest (MD5 hash)**
4) **Appends** that digest to the **OSPF packet**
5) **Sends** the packet

The destination router…

1) Receives the packet and the **attached MD5 digest**
2) Uses its copy of the **same shared key**
3) **Computes** its own **MD5 hash** for the **received packet**
4) **Compares** its **computed hash** to the **received digest**
5) **Accepts the packet** only if they **match**

Type 2 authentication also introduces additional features not available with Type 1, including the use of **key IDs and sequence numbers**, which allow for smoother key changes without disrupting neighbor adjacencies. While MD5 is no longer considered a strong cryptographic algorithm by modern security standards, it remains widely supported and is useful for understanding how cryptographic authentication works in OSPFv2.

Step 2: Remove OSPFv2 Type 1 Cleartext Authentication and Verify Adjacency

a. Before configuring OSPFv2 Type 2 (MD5) authentication, the existing **Type 1 cleartext authentication** must be removed. This ensures that both routers start from a consistent, unauthenticated state and prevents authentication mismatches during the transition.

```
R1(config)# interface g0/0/1
R1(config-if)# no ip ospf authentication-key PLAINTXT
R1(config-if)# no ip ospf authentication
R1(config-if)# exit
*Jan 19 19:21:11.579: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.2 on GigabitEthernet0/0/1
from FULL to DOWN, Neighbor Down: Dead timer expired


D1(config)# interface g1/0/11
```

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```
D1(config-if)# no ip ospf authentication-key PLAINTXT
D1(config-if)# exit
D1(config)# router ospf 123
D1(config-router)# no area 1 authentication
D1(config-router)# exit
*Jan 19 19:17:10.385: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.1 on
GigabitEthernet1/0/11 from LOADING to FULL, Loading Done
```

b. Verify that the neighbor adjacency between R1 and D1 has successfully re-established.

```
R1# show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.2           0   FULL/  -        00:00:33    172.16.2.2      GigabitEthernet0/0/1

D1# show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:34    172.16.2.1      GigabitEthernet1/0/11
```

Step 3: On R1, enable MD5 authentication and define the key on the link.

When deploying OSPF MD5 authentication, you need to configure the key ID and the key (password), by entering the **ip ospf message-digest-key** command. Cisco IOS Software will give you a warning if you try to configure a password that is longer than 16 characters; only the first 16 characters will be used.

On R1, enable OSPF MD5 authentication on the interface connecting to D1 and configure the shared MD5 key. If you're not quick enough, you will see a warning NOVALIDKEY message before you configure the key. Also, note that adjacency with D1 transitions to DOWN after the dead timer expires.

```
R1(config)# interface g0/0/1
R1(config-if)# ip ospf authentication message-digest
*Jan 19 19:29:56.382: %OSPF-4-NOVALIDKEY: No valid authentication send key is
available on interface GigabitEthernet0/0/1
R1(config-if)# ip ospf message-digest-key 1 md5 secretkey
R1(config-if)# exit
R1(config)#
*Jan 19 19:30:31.255: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.2 on GigabitEthernet0/0/1
from FULL to DOWN, Neighbor Down: Dead timer expired
```

Command Summary:

- **message-digest** enables cryptographic authentication on the interface.
- **message-digest-key 1** defines **key ID 1**, and **secretkey** is the **shared MD5 secret** used to generate the message digest.
- The key itself is never sent in cleartext.

Step 4: On D1, configure the MD5 key on the link and enable MD5 authentication in OSPFv2 for Area 1.

Configure the **same MD5 key and key ID** on the interface connecting D1 to R1 and enable MD5 authentication at the **OSPF area level** on D1. After configuring authentication, adjacency with R1 is reestablished.

```
D1(config)# interface g1/0/11
```

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```
D1(config-if)# ip ospf message-digest-key 1 md5 secretkey
D1(config-if)# exit
```

Important: The key ID and key value must match exactly on both routers.

```
D1(config)# router ospf 123
D1(config-router)# area 1 authentication message-digest
D1(config-router)# exit
*Jan 19 19:29:07.987: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.1 on
GigabitEthernet1/0/11 from LOADING to FULL, Loading Done
```

Applying authentication at the area level ensures that all OSPF-enabled interfaces in Area 1 require MD5 authentication unless overridden on a specific interface.

Resolve the MD5 Authentication Warning on D1.

After enabling **OSPFv2 Type 2 (MD5) authentication** at the area level, you probably saw the following message on **D1**:

```
*Jan 19 19:30:13.727: %OSPF-4-NOVALIDKEY: No valid authentication send key is
available on interface GigabitEthernet1/0/23
```

Explanation

This message indicates that **OSPF MD5 authentication is enabled on an interface that does not have a valid MD5 key configured**. When authentication is applied at the area level, **all OSPF-enabled interfaces in that area are expected to use MD5 authentication**.

On D1, the MD5 key was configured only on the interface connecting to **R1 (GigabitEthernet1/0/11)**. However, the LAN-facing interface **GigabitEthernet1/0/23** is also participating in OSPF Area 1 and does not have an MD5 key configured. As a result, OSPF is unable to send authenticated packets on that interface and generates the warning message.

Solution

Configure the LAN interface on D1 as a passive interface. This prevents OSPF from sending or receiving routing protocol packets on the LAN while still allowing the network to be advertised into OSPF.

```
D1(config)# router ospf 123
D1(config-router)# passive-interface GigabitEthernet1/0/23
```

The OSPF neighbor adjacency with R1 should remain in the **FULL** state, and the authentication warning message should no longer appear.

Note

As an alternative to configuring the LAN interface as passive, an MD5 authentication key **could also be configured on interface GigabitEthernet1/0/23**. This would satisfy the requirement for area-level authentication, since all OSPF-enabled interfaces in the area must have a valid authentication key when MD5 is used. However, because this interface does not form OSPF neighbor relationships, configuring it as a passive interface is the preferred and more common practice.

This behavior may not have been noticeable when using **Type 1 (cleartext) authentication**. Type 1 authentication is enforced less strictly by IOS, and missing authentication keys on interfaces that do not actively exchange OSPF packets may not generate explicit warning messages.

If a similar issue were encountered with cleartext authentication, the **same solution, configuring the interface as passive, would apply**.

Understanding MD5 Authentication, Key IDs, and Key Rollover

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The configured key (for example, **secretkey**) and key ID (for example, **1**) are used to generate a message digest (hash) for each OSPF packet. This digest is appended to the packet and allows neighboring routers to verify the authenticity and integrity of OSPF messages.

Typically, a single key is used per interface. All neighboring routers on the same network must be configured with the **same key ID and key value** to form an OSPF adjacency.

The use of a key ID allows for uninterrupted key changes. During a key transition, a router can be configured with multiple keys and will temporarily send duplicate OSPF packets, each authenticated with a different key. Once all neighbors adopt the new key, the router automatically stops sending packets authenticated with the old key.

This mechanism allows administrators to change OSPF authentication keys without disrupting routing adjacencies.

Step 5: Verify OSPFv2 neighbor adjacency between R1 and D1 after authentication

After configuring OSPFv2 authentication, verify that the neighbor adjacency between R1 and D1 has successfully re-established.

```
R1# show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.2           0   FULL/  -        00:00:39    172.16.2.2      GigabitEthernet0/0/1

D1# show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:35    172.16.2.1      GigabitEthernet1/0/11
```

Step 6: Verify OSPFv2 Type 2 Authentication on the Interface

Use the **show ip ospf interface** command to confirm that **OSPFv2 Type 2 (MD5) authentication is enabled on the interface**. The line **Cryptographic authentication enabled** and the presence of **key ID 1** indicate that MD5 authentication is active and being used to authenticate OSPF packets exchanged with the neighbor.

```
D1# show ip ospf interface g 1/0/11
GigabitEthernet1/0/11 is up, line protocol is up (connected)
  Internet Address 172.16.2.2/30, Interface ID 40, Area 1
  Attached via Interface Enable
  Process ID 123, Router ID 1.1.1.2, Network Type POINT_TO_POINT, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Enabled by interface config, including secondary ip addresses
  Transmit Delay is 1 sec, State POINT_TO_POINT
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  <output omitted>
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.1
  Suppress hello for 0 neighbor(s)
  Cryptographic authentication enabled
    Youngest key id is 1
```

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Part 5: OSPFv3 IPsec AH (Authentication & Integrity)

In this part, you will configure **OSPFv3 authentication using IPsec Authentication Header (AH)**. This method protects OSPFv3 routing exchanges by ensuring that routing packets are **authentic** and have **not been altered in transit**, while leaving the packet contents unencrypted.

Why OSPFv3 Uses IPsec

Unlike OSPFv2, OSPFv3 no longer includes authentication fields in the OSPF packet header. Instead, OSPFv3 relies on **IPsec-based security mechanisms** to protect routing updates. This applies regardless of whether OSPFv3 is used to carry IPv6 routes or IPv4 routes through OSPFv3 address families.

OSPFv3 packets are transported using IPv6, and security is provided through **IPv6 extension headers**, specifically **IPsec Authentication Header (AH)** and **IPsec Encapsulating Security Payload (ESP)**.

This design reflects a major architectural shift: *OSPFv3 delegates security to the network layer rather than embedding it directly into the routing protocol.*

Note: Even when OSPFv3 is used for IPv4 routing, OSPFv3 packets are transported using IPv6. Therefore, OSPFv3 authentication and security are always implemented using IPv6 mechanisms such as IPsec.

What IPsec AH Provides

When IPsec AH is used with OSPFv3, it provides:

- **Authentication** - verifies that routing packets are sent by a trusted neighbor
- **Integrity** - ensures that packets have not been modified in transit

IPsec AH does **not** provide **encryption**. OSPF packets remain **visible on the wire**, but any unauthorized modification or spoofing attempt will be detected and rejected.

This protection helps prevent common routing attacks, such as:

- Injection of rogue routing updates
- Formation of unauthorized OSPF neighbor relationships

Security Policies in OSPFv3

To use IPsec AH with OSPFv3, a **security policy** must be defined. A policy consists of:

- A **Security Parameter Index (SPI)**, which identifies the security association
- A **shared key**, used to generate and validate cryptographic hashes

These parameters must match on neighboring routers for OSPFv3 adjacencies to form.

IPsec policies for OSPFv3 can be applied two ways:

- **Per interface**, providing the highest level of control and security
- **Per OSPFv3 area**, where the policy applies to all interfaces in that area unless overridden

Once configured, the IPsec process operates **transparently**. OSPFv3 continues to function normally, while IPsec protects the routing exchanges behind the scenes.

Important Implementation Notes

- OSPFv3 authentication using IPsec requires a **crypto-capable IOS image**, as IPsec functionality is needed by the routing process.
- IPsec AH is enabled using the **ospfv3 authentication** command.

Next, you will configure **OSPFv3 authentication using IPsec Authentication Header (AH)** on the link between **R1 and R2**. Authentication will be applied **only to the connecting interface**, ensuring that other OSPFv3 links are unaffected. The same authentication policy will protect **both IPv6 and IPv4 address families**.

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Step 1: Verify OSPFv3 neighbor adjacencies between D1 and D2.

a. On D1, verify the OSPFv3 neighbor adjacency with D2.

```
D1# show ospfv3 neighbor

          OSPFv3 123 address-family ipv4 (router-id 1.1.1.2)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
1.1.1.3           0   FULL/  -        00:00:31    9               GigabitEthernet1/0/1

          OSPFv3 123 address-family ipv6 (router-id 1.1.1.2)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
1.1.1.3           0   FULL/  -        00:00:38    9               GigabitEthernet1/0/1
```

b. On D2, verify the OSPFv3 neighbor adjacency with D1.

```
D2# show ospfv3 neighbor

          OSPFv3 123 address-family ipv4 (router-id 1.1.1.3)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
1.1.1.2           0   FULL/  -        00:00:37    9               GigabitEthernet1/0/1

          OSPFv3 123 address-family ipv6 (router-id 1.1.1.3)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
1.1.1.2           0   FULL/  -        00:00:31    9               GigabitEthernet1/0/1
```

Step 2: Verify OSPFv3 reachability to remote IPv4 and IPv6 networks.

a. On D1, use the **show ip route ospfv3** command to verify reachability to D2's IPv4 172.16.10.0/24 network.

```
D1# show ip route ospfv3 | begin Gateway
Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 7 subnets, 3 masks
O        172.16.10.0/24 [110/2] via 172.16.1.2, 01:05:34, GigabitEthernet1/0/1
D1#
```

b. On D1, use the **show ipv6 route ospf** command to verify reachability to D2's IPv6 2001:db8:acad:a010::/64 network.

```
D1# show ipv6 route ospf | begin 2001
O   2001:DB8:ACAD:A010::/64 [110/2]
     via FE80::D2:1, GigabitEthernet1/0/1 D1#
```

Step 3: Configure OSPFv3 IPsec AH authentication on D1

On **D1**, enable OSPFv3 IPsec AH authentication on the interface connecting to **D2**. Use a shared key of your choice. The SPI value, authentication algorithm, and key **must match** on both routers. For SHA-1, the key length must be exactly 40 hexadecimal characters.

```
D1(config)# interface GigabitEthernet1/0/1
```

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```
D1(config-if)# ospfv3 authentication ipsec spi 256 sha1
1234567890123456789012345678901234567890
*Jan 19 19:47:13.692: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
```

Why 40 hexadecimal characters? SHA-1 uses a 160-bit key. Since each hexadecimal character represents 4 bits, a total of 40 hexadecimal characters is required (160 ÷ 4 = 40).

Authentication is applied per interface, ensuring that only the D1–D2 link is protected.

After enabling authentication on D1, you will observe that the OSPFv3 neighbor adjacency for both IPv4 and IPv6 temporarily drops due to an authentication mismatch, since D2 has not yet been configured.

```
*Jan 19 19:47:47.019: %OSPFv3-5-ADJCHG: Process 123, IPv4, Nbr 1.1.1.3 on
GigabitEthernet1/0/1 from FULL to DOWN, Neighbor Down: Dead timer expired
*Jan 19 19:47:51.939: %OSPFv3-5-ADJCHG: Process 123, IPv6, Nbr 1.1.1.3 on
GigabitEthernet1/0/1 from FULL to DOWN, Neighbor Down: Dead timer expired
```

Step 4: Configure OSPFv3 IPsec AH authentication on D2.

Next, configure the **same IPsec AH authentication parameters** on the interface connecting **D2 to D1**.

```
D2(config)# interface GigabitEthernet1/0/1
D2(config-if)# ospfv3 authentication ipsec spi 256 sha1
1234567890123456789012345678901234567890
*Jan 19 18:31:11.307: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
*Jan 19 18:31:11.742: %OSPFv3-5-ADJCHG: Process 123, IPv6, Nbr 1.1.1.2 on
GigabitEthernet1/0/1 from LOADING to FULL, Loading Done
*Jan 19 18:31:15.738: %OSPFv3-5-ADJCHG: Process 123, IPv4, Nbr 1.1.1.2 on
GigabitEthernet1/0/1 from LOADING to FULL, Loading Done
```

Once authentication is configured on D2, the OSPFv3 neighbor adjacency between D1 and D2 should re-establish for **both IPv6 and IPv4 address families**.

Both routers must use:

- The same SPI value
- The same authentication algorithm
- The same key

To apply IPsec authentication to an entire area instead, the configuration would be done under the OSPFv3 router process using the **area *area-id* authentication ipsec spi** command.

Note: When using OSPFv3 with IPsec AH or ESP, SHA-1 is the most commonly supported authentication algorithm across Cisco IOS-XE platforms. Stronger SHA-based algorithms are supported when using the OSPFv3 Authentication Trailer (RFC 7166), which uses key chains instead of IPsec.

Step 5: Verify neighbor adjacency and reachability after OSPFv3 IPsec AH authentication is configured.

a. Verify D1's neighbor adjacency was re-established with D2.

```
D1# show ospfv3 neighbor

          OSPFv3 123 address-family ipv4 (router-id 1.1.1.2)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
1.1.1.3           0   FULL/  -        00:00:38    9               GigabitEthernet1/0/1

          OSPFv3 123 address-family ipv6 (router-id 1.1.1.2)
```

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```
Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
1.1.1.3           0   FULL/  -        00:00:37    9               GigabitEthernet1/0/1
```

b. Verify D1's reachability to the remote IPv4 network on D2.

```
D1# show ip route ospfv3 | begin Gateway
Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 7 subnets, 3 masks
O        172.16.10.0/24 [110/2] via 172.16.1.2, 01:11:00, GigabitEthernet1/0/1
```

c. Verify D1's reachability to the remote IPv6 network on D2.

```
D1# show ipv6 route ospf | begin 2001
O   2001:DB8:ACAD:A010::/64 [110/2]
     via FE80::D2:1, GigabitEthernet1/0/1
```

Step 6: Verify OSPFv3 IPsec AH authentication using the show crypto ipsec sa command.

The **show crypto ipsec sa** command confirms that IPsec Authentication Header (AH) is actively protecting OSPFv3 routing exchanges on the D1–D2 link.

```
D1# show crypto ipsec sa

interface: GigabitEthernet1/0/1
    Crypto map tag: GigabitEthernet1/0/1-OSPF-MAP, local addr FE80::D1:1

   IPsecv6 policy name: OSPFv3-256

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (FE80::/10/89/0)
   remote ident (addr/mask/prot/port): (::/0/89/0)
   current_peer FF02::5 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 143, #pkts encrypt: 143, #pkts digest: 143
    #pkts decaps: 143, #pkts decrypt: 143, #pkts verify: 143
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: FE80::D1:1,
     remote crypto endpt.: FF02::5
     plaintext mtu 1476, path mtu 1500, ipv6 mtu 1500, ipv6 mtu idb
GigabitEthernet1/0/1
     current outbound spi: 0x100(256)
     PFS (Y/N): N, DH group: none

     inbound esp sas:

     inbound ah sas:
      spi: 0x100(256)
        transform: ah-sha-hmac ,
```

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```
        in use settings ={Transport, }
        conn id: 1, flow_id: SW:1, sibling_flags FFFFFFFF80000011, crypto map:
GigabitEthernet1/0/1-OSPF-MAP, initiator : False
        sa timing: remaining key lifetime (sec): 0
        Kilobyte Volume Rekey has been disabled
        replay detection support: N
        Status: ACTIVE(ACTIVE)

     inbound pcp sas:

     outbound esp sas:

     outbound ah sas:
      spi: 0x100(256)
        transform: ah-sha-hmac ,
        in use settings ={Transport, }
        conn id: 2, flow_id: SW:2, sibling_flags FFFFFFFF80000011, crypto map:
GigabitEthernet1/0/1-OSPF-MAP, initiator : False
        sa timing: remaining key lifetime (sec): 0
        Kilobyte Volume Rekey has been disabled
        replay detection support: N
        Status: ACTIVE(ACTIVE)

     outbound pcp sas:
```

Examining the output above:

- **IPsecv6 policy name: OSPFv3-256** - This confirms that an IPsec policy created specifically for OSPFv3 is applied to the interface.
- **#pkts digest: 143**, **#pkts verify: 143**, and **#send errors 0, #recv errors 0** - These counters show that OSPFv3 packets are being successfully authenticated and verified, with no authentication failures.
- **local crypto endpt.: FE80::D1:1** and **remote crypto endpt.: FF02::5** - This confirms that OSPFv3 control traffic is protected over IPv6 using the OSPFv3 multicast address (FF02::5), consistent with OSPFv3 operation.
- **current outbound spi: 0x100(256)** - The SPI value matches the configured SPI (256), verifying that the correct security association is being used.
- **transform: ah-sha-hmac** - This confirms that IPsec AH is in use and that packets are being authenticated and integrity-protected using SHA-based HMAC.

This output confirms that IPsec AH is actively protecting OSPFv3 routing traffic on the D1–D2 link. The presence of active inbound and outbound AH security associations, matching SPI values, SHA-based authentication, and successful packet verification counters demonstrates that OSPFv3 authentication and integrity checking are functioning as intended.

Step 7: Verify OSPFv3 IPsec AH Authentication using the show ospfv3 interface command.

The **show ospfv3 interface** command provides a protocol-level view of OSPFv3 operation on an interface and is useful for confirming that IPsec authentication is enabled and functioning. In this output, D1 displays the OSPFv3 status for **both address families** on the same interface, allowing us to verify that authentication is applied consistently.

```
D1# show ospfv3 interface g 1/0/1
GigabitEthernet1/0/1 is up, line protocol is up (connected)
```

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```
  Link Local Address FE80::D1:1, Interface ID 9 (snmp-if-index)
  Internet Address 172.16.1.1/30
  Area 0, Process ID 123, Instance ID 64, Router ID 1.1.1.2
  Network Type POINT_TO_POINT, Cost: 1
  SHA-1 authentication SPI 256, secure socket UP (errors: 0)
  Transmit Delay is 1 sec, State POINT_TO_POINT
  <output omitted>
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.3
  Suppress hello for 0 neighbor(s)
GigabitEthernet1/0/1 is up, line protocol is up (connected)
  Link Local Address FE80::D1:1, Interface ID 9 (snmp-if-index)
  Area 0, Process ID 123, Instance ID 0, Router ID 1.1.1.2
  Network Type POINT_TO_POINT, Cost: 1
  SHA-1 authentication SPI 256, secure socket UP (errors: 0)
  Transmit Delay is 1 sec, State POINT_TO_POINT
  <output omitted>
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.3
  Suppress hello for 0 neighbor(s)
D1#
```

The output appears twice because OSPFv3 is running with address families. The first instance represents the **IPv4 address family**, indicated by the **presence of the IPv4 interface address (172.16.1.1/30)** and Instance ID 64. In this case, OSPFv3 is being used to advertise IPv4 routes, even though the protocol itself still operates over IPv6. The second instance represents the **IPv6 address family**, where OSPFv3 exchanges IPv6 routing information using only IPv6 addressing and instance ID 0. This IPv6 instance only has an IPv6 link-local address. Both instances contain the IPv6 link-local address because OSPFv3 uses IPv6 to exchange routing information.

The important line confirming successful authentication is shown in **both sections** of the output:

```
SHA-1 authentication SPI 256, secure socket UP (errors: 0)
```

This line indicates that:

- **SHA-1 authentication** is being used with **IPsec AH**
- The configured **SPI value (256)** matches the authentication policy
- The **secure socket is UP**, meaning authentication has been successfully established
- **No authentication errors** have occurred

Part 6: OSPFv3 IPsec ESP (encryption + authentication)

In the previous section, you secured OSPFv3 routing updates using **IPsec AH**, which provides **authentication and integrity** but does **not encrypt** the contents of OSPF packets. While AH ensures that routing updates come from a trusted neighbor and have not been altered in transit, the routing information itself remains visible to anyone who can capture the packets on the link.

In this part, you will extend that security model by configuring **IPsec ESP (Encapsulating Security Payload)** for OSPFv3. ESP builds on the protection offered by AH by adding **confidentiality through encryption**, while still providing authentication and integrity checks. This means OSPFv3 routing updates are not only verified and protected from tampering, but are also **encrypted**, preventing unauthorized devices from viewing routing information.

How ESP Is Similar to IPsec AH

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From a configuration and operational perspective, IPsec ESP behaves similarly to IPsec AH in several important ways:

- Both mechanisms rely on **IPsec security associations (SAs)**.
- Both require a **Security Parameters Index (SPI)** and a **shared key**.
- Both can be applied **per interface or per OSPFv3 area**.
- Both protect OSPFv3 traffic carried over **IPv6**, regardless of whether IPv4 or IPv6 routes are being advertised.

As with AH, **all neighboring routers must be configured with matching parameters** (SPI, algorithms, and keys) to form and maintain OSPF adjacencies.

How ESP Differs from IPsec AH

The key difference is that **ESP encrypts the OSPFv3 payload**, whereas AH does not.

With IPsec **AH**:

- **Routing updates are authenticated**, ensuring they originate from a trusted neighbor.
- **Routing updates are protected from modification**, providing integrity.
- **Routing updates are not encrypted** and remain visible on the wire

Note: While AH itself does not encrypt packets, routing updates could still be encrypted if the link is protected by an external VPN or tunnel.

With IPsec **ESP**:

- **Routing updates are authenticated**, ensuring they originate from a trusted neighbor.
- **Routing updates are protected from modification**, providing integrity.
- **Routing updates are encrypted**, providing confidentiality and hiding packet contents from capture.

Because ESP provides stronger protection, it is typically preferred in environments where **routing information confidentiality is important**.

Security Parameters and SPI Usage

When configuring OSPFv3 IPsec ESP, each interface must use a **unique SPI value**. The SPI identifies the security association used to protect packets on that link. In this lab, the SPI used between one pair of neighbors is intentionally different from the SPI used on other links to reinforce this best practice.

If ESP parameters are not correctly configured on both ends of the link, the OSPFv3 adjacency will fail after the dead timer expires. Once matching ESP settings are applied on both routers, the adjacency is automatically re-established for **both IPv4 and IPv6 address families**.

Supported ESP and AH Algorithm Options

The following tables summarize the encryption and authentication options available when configuring OSPFv3 using IPsec ESP and AH:

Note: All key lengths shown below are expressed as **hexadecimal strings**. **1 hex character = 4 bits**

Encryption Algorithms (ESP)

The table below summarizes the **encryption algorithms supported by IPsec ESP when used with OSPFv3**, along with their required key sizes. ESP provides **confidentiality** by encrypting OSPFv3 routing updates, preventing unauthorized devices from viewing routing information. When configuring ESP, the encryption key must be entered as a **hexadecimal string of a specific length**, which depends on the selected algorithm and key size.

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Encryption Algorithm (ESP)	Key Size (bits)	Required Key Length (Hex Characters)	Notes
DES	64-bit	16 characters	Legacy, not recommended
3DES	192-bit	48 characters	Legacy, stronger than DES
AES-CBC	128-bit	32 characters	Common, recommended, used in this lab
AES-CBC	192-bit	48 characters	Stronger encryption
AES-CBC	256-bit	64 characters	Strongest option

Authentication Algorithms

The table below lists the **authentication algorithms supported by IPsec AH for OSPFv3**, along with their required key lengths. AH provides **authentication and integrity**, ensuring that OSPFv3 routing updates originate from trusted neighbors and have not been modified in transit. Like ESP, AH requires keys to be specified as **hexadecimal strings**, and the length of the key must exactly match the requirements of the selected algorithm.

Authentication Algorithm (AH)	Hash Size (bits)	Required Key Length (Hex Characters)	Notes
MD5	128-bit	32 characters	Legacy, weak
SHA-1	160-bit	40 characters	Used in this lab
SHA-256 *(platform dependent)*	256-bit	64 characters	More secure, may not be available on all devices

Step 1: Remove OSPFv3 IPsec AH configurations on D1 and D2.

On both D1 and D2, remove the IPsec AH configurations. If you do it fast enough, neighbor adjacency should remain FULL.

```
D1(config)# interface GigabitEthernet1/0/1
D1(config-if)# no ospfv3 authentication ipsec spi 256 sha1
12345678901234567890123456789012345678901234567890
*Jan 19 19:00:14.796: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF


D2(config)# interface GigabitEthernet1/0/1
D2(config-if)# no ospfv3 authentication ipsec spi 256 sha1
12345678901234567890123456789012345678901234567890
*Jan 19 20:18:01.993: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
```

Step 2: Configure OSPFv3 IPsec ESP Encryption Between D1 and D2

In this step, you will secure OSPFv3 routing updates between **D1 and D2** using **IPsec ESP**, which provides **encryption, authentication, and integrity**. The ESP configuration is applied **only to the directly connected link** between the two devices so that other OSPFv3 interfaces are not affected.

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Just as in the previous AH-based configuration, both devices must be configured with:

- The **same SPI**
- The **same encryption algorithm and key**
- The **same authentication algorithm and key**

A **unique SPI** should be used per secured link to avoid conflicts with other OSPFv3 IPsec associations.

a. Configure ESP on D1.

On **D1**, enable OSPFv3 IPsec ESP encryption on the interface connecting to D2.

Note: The format of this lab shows the **ospfv3 encryption ipsec spi** command on two lines. The command **must be entered as a single uninterrupted line**.

```
D1(config)# interface GigabitEthernet1/0/1
D1(config-if)# ospfv3 encryption ipsec spi 257 esp aes-cbc 128
12345678901234567890123456789012 sha1
1234567890123456789012345678901234567890
*Jan 19 20:20:54.860: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
*Jan 19 20:21:26.667: %OSPFv3-5-ADJCHG: Process 123, IPv4, Nbr 1.1.1.3 on
GigabitEthernet1/0/1 from FULL to DOWN, Neighbor Down: Dead timer expired
*Jan 19 20:21:30.023: %OSPFv3-5-ADJCHG: Process 123, IPv6, Nbr 1.1.1.3 on
GigabitEthernet1/0/1 from FULL to DOWN, Neighbor Down: Dead timer expired
```

Note: After the dead timer expires, OSPFv3 adjacency will drop because encryption is enabled on D1 but not yet configured on D2.

Explanation:

- The **ospfv3 encryption ipsec spi** command enables **ESP encryption** for OSPFv3 on a per-interface basis.
- **SPI 257** is used to uniquely identify the IPsec security association for OSPFv3 encryption on this interface and must match on both neighbors.
- **AES-CBC 128-bit** encrypts the OSPFv3 payload, providing confidentiality and must match on both neighbors.
- **SHA-1** ensures authentication and integrity of routing updates and must match on both neighbors.
- Because OSPFv3 is running with **address families**, the adjacency is re-established for **both IPv4 and IPv6**, even though the protocol itself operates over IPv6.
- Applying the configuration at the **interface level** ensures that **only the intended link is secured**.

b. Configure ESP on D2.

Next, apply the **same ESP configuration** on **D2** for the link back to D1.

```
D2(config)# interface GigabitEthernet1/0/1
D2(config-if)# ospfv3 encryption ipsec spi 257 esp aes-cbc 128
12345678901234567890123456789012 sha1
1234567890123456789012345678901234567890
*Jan 19 19:04:51.558: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
*Jan 19 19:04:52.703: %OSPFv3-5-ADJCHG: Process 123, IPv4, Nbr 1.1.1.2 on
GigabitEthernet1/0/1 from LOADING to FULL, Loading Done
*Jan 19 19:04:58.478: %OSPFv3-5-ADJCHG: Process 123, IPv6, Nbr 1.1.1.2 on
GigabitEthernet1/0/1 from LOADING to FULL, Loading Done
```

Note: Once the configuration matches on both devices, the OSPFv3 adjacency is automatically re-established.

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Step 3: Verify neighbor adjacency and reachability after OSPFv3 IPsec ESP authentication is configured.

a. Verify D1's neighbor adjacency was re-established with D2.

```
D1# show ospfv3 neighbor

          OSPFv3 123 address-family ipv4 (router-id 1.1.1.2)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
1.1.1.3           0   FULL/  -        00:00:33    9               GigabitEthernet1/0/1

          OSPFv3 123 address-family ipv6 (router-id 1.1.1.2)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
1.1.1.3           0   FULL/  -        00:00:35    9               GigabitEthernet1/0/1
```

b. Verify D1's reachability to the remote IPv4 network on D2.

```
D1# show ip route ospfv3 | begin Gateway
Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 7 subnets, 3 masks
O        172.16.10.0/24 [110/2] via 172.16.1.2, 04:04:43, GigabitEthernet1/0/1
D1#
```

c. Verify D1's reachability to the remote IPv6 network on D2.

```
D1# show ipv6 route ospf | begin 2001
O   2001:DB8:ACAD:A010::/64 [110/2]
     via FE80::D2:1, GigabitEthernet1/0/1
```

Step 4: Verify OSPFv3 IPsec AH Authentication using the show crypto ipsec sa command.

The **show crypto ipsec sa** command confirms that **IPsec ESP** is actively protecting OSPFv3 routing exchanges on the **D1–D2 link**. Unlike IPsec AH, ESP provides **encryption, authentication, and integrity**, ensuring that OSPFv3 routing updates are confidential, authenticated, and protected from modification.

```
D1# show crypto ipsec sa

interface: GigabitEthernet1/0/1
    Crypto map tag: GigabitEthernet1/0/1-OSPF-MAP, local addr FE80::D1:1

   IPsecv6 policy name: OSPFv3-257

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (FE80::/10/89/0)
   remote ident (addr/mask/prot/port): (::/0/89/0)
   current_peer FF02::5 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 108, #pkts encrypt: 108, #pkts digest: 108
    #pkts decaps: 107, #pkts decrypt: 107, #pkts verify: 107
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
```

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```
      #send errors 0, #recv errors 0

       local crypto endpt.: FE80::D1:1,
       remote crypto endpt.: FF02::5
       plaintext mtu 1462, path mtu 1500, ipv6 mtu 1500, ipv6 mtu idb
GigabitEthernet1/0/1
       current outbound spi: 0x101(257)
       PFS (Y/N): N, DH group: none

       inbound esp sas:
        spi: 0x101(257)
          transform: esp-aes esp-sha-hmac ,
          in use settings ={Transport, }
          conn id: 3, flow_id: SW:3, sibling_flags FFFFFFFF80000001, crypto map:
GigabitEthernet1/0/1-OSPF-MAP, initiator : False
          sa timing: remaining key lifetime (sec): 0
          Kilobyte Volume Rekey has been disabled
          IV size: 16 bytes
          replay detection support: N
          Status: ACTIVE(ACTIVE)

       inbound ah sas:

       inbound pcp sas:

       outbound esp sas:
        spi: 0x101(257)
          transform: esp-aes esp-sha-hmac ,
          in use settings ={Transport, }
          conn id: 4, flow_id: SW:4, sibling_flags FFFFFFFF80000001, crypto map:
GigabitEthernet1/0/1-OSPF-MAP, initiator : False
          sa timing: remaining key lifetime (sec): 0
          Kilobyte Volume Rekey has been disabled
          IV size: 16 bytes
          replay detection support: N
          Status: ACTIVE(ACTIVE)

       outbound ah sas:

       outbound pcp sas:
```

Examining the output above:

- **IPsecv6 policy name: OSPFv3-257** - This confirms that an IPsec policy created specifically for **OSPFv3 ESP** is applied to the interface. The policy name reflects the configured SPI value (257).
- **#pkts encaps** / **encrypt** / **digest** and **#pkts decaps** / **decrypt** / **verify** - These counters show that OSPFv3 packets are being successfully **encrypted, authenticated, decrypted, and verified**. The absence of send or receive errors confirms that ESP is functioning correctly on the link.
- **local crypto endpt.: FE80::D1:1** and **remote crypto endpt.: FF02::5** - This confirms that OSPFv3 control traffic is protected over IPv6 using **link-local addresses** and the OSPFv3 **multicast address** (FF02::5), which is consistent with OSPFv3 operation.

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- **current outbound spi: 0x101 (257** - The SPI value matches the configured SPI (257), verifying that the correct **security association** is being used for ESP encryption.
- **inbound esp sas** / **outbound esp sas** - The presence of active inbound and outbound **ESP security associations** confirms that ESP (not AH) is protecting the OSPFv3 packets on this interface.
- **transform: esp-aes esp-sha-hmac** - This confirms that **AES** is being used to encrypt the OSPFv3 payload and **SHA-based HMAC** is being used to authenticate and verify packet integrity.
- **Status: ACTIVE(ACTIVE)** - Indicates that the ESP security associations are currently active and in use.

Step 5: Verify OSPFv3 IPsec ESP Authentication using the show ospfv3 interface command.

The show ospfv3 interface command provides confirmation that **IPsec ESP encryption** is enabled and operational on the OSPFv3 interface. This output is similar to what was observed with IPsec AH; however, it now reflects the use of **encryption in addition to authentication and integrity**.

```
D1# show ospfv3 interface g 1/0/1
GigabitEthernet1/0/1 is up, line protocol is up (connected)
  Link Local Address FE80::D1:1, Interface ID 9 (snmp-if-index)
  Internet Address 172.16.1.1/30
  Area 0, Process ID 123, Instance ID 64, Router ID 1.1.1.2
  Network Type POINT_TO_POINT, Cost: 1
  AES-CBC-128 encryption SHA-1 auth SPI 257, secure socket UP (errors: 0)
  <output omitted>
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.3
  Suppress hello for 0 neighbor(s)
GigabitEthernet1/0/1 is up, line protocol is up (connected)
  Link Local Address FE80::D1:1, Interface ID 9 (snmp-if-index)
  Area 0, Process ID 123, Instance ID 0, Router ID 1.1.1.2
  Network Type POINT_TO_POINT, Cost: 1
  AES-CBC-128 encryption SHA-1 auth SPI 257, secure socket UP (errors: 0)
  <output omitted>
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.3
  Suppress hello for 0 neighbor(s)
```

Examine the following line:

```
AES-CBC-128 encryption SHA-1 auth SPI 257
```

This confirms that IPsec ESP is in use, with:

- **AES-CBC 128-bit** providing encryption (confidentiality)
- **SHA-1** providing authentication and integrity
- **SPI 257** identifying the active IPsec security association in the AH configuration, this line referenced only authentication (no encryption).

Note: As with previous OSPFv3 verification steps, the output appears twice—once for the IPv4 address family (Instance ID 64) and once for the IPv6 address family (Instance ID 0). Although OSPFv3 is advertising both IPv4 and IPv6 routes, the protocol—and therefore IPsec—continues to operate over IPv6, using the interface's link-local address.

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Part 7: OSPFv3 Authentication Trailer (RFC 7166)

Up to this point in the lab, you have secured OSPFv3 using **IPsec AH and IPsec ESP**, which operate at the IPv6 layer and protect OSPFv3 packets using **standard IPsec security associations**. This reflects the original design philosophy of OSPFv3: *remove authentication from the routing protocol itself and rely on IPsec for security*.

However, in real networks, this approach is not always practical.

Configuring and maintaining IPsec, especially for routing protocol traffic, can introduce operational complexity. Some platforms, images, or deployment environments may not fully support IPsec for OSPFv3 (as is the case with some Cisco IOS-XE router platforms), or administrators may prefer a lighter-weight solution that more closely resembles the authentication model used in OSPFv2.

To address this gap, **RFC 7166** introduces the **OSPFv3 Authentication Trailer**, a protocol-level authentication mechanism that provides cryptographic protection ***without* using IPsec**.

The OSPFv3 Authentication Trailer is not part of the OSPFv3 packet header; it is a separate trailer appended to the end of the packet and is indicated by the AT-bit in the OSPFv3 Options field, which is part of the OSPFv3 Header.

Encapsulation is as follows:

- **OSPFv3 Header** (includes Options field with AT-bit set)
- **OSPFv3 Payload**
- **Authentication Trailer** (HMAC, etc.)

It offers a simpler operational model than IPsec while still protecting OSPFv3 from spoofing and tampering—making it an important tool in the OSPFv3 security toolbox.

What the Authentication Trailer Is and Is Not

The OSPFv3 Authentication Trailer adds a **cryptographic authentication block to the end of each OSPFv3 packet**. This trailer contains an **HMAC (Hashed Message Authentication Code)** calculated using a shared secret and a secure hash algorithm (such as **SHA-256**).

It provides:

- **Authentication** - verifies that OSPF packets come from a trusted neighbor
- **Integrity** - ensures that packets have not been modified in transit

It does **not** provide:

- **Encryption** - OSPFv3 packet contents remain visible on the wire

OSPFv3 Authentication Trailer (RFC 7166) provides cryptographic authentication similar in purpose to OSPFv2 Type 2 (MD5) authentication but uses modern HMAC-SHA algorithms and a trailer-based design that aligns with OSPFv3's architecture.

Why This Exists Alongside IPsec

The Authentication Trailer is **not a replacement for IPsec**. Instead, it is an **alternative** designed for environments where:

- **IPsec is difficult to deploy** or operationally heavy.
- Simpler **protocol-level authentication** is preferred.
- **Strong cryptographic authentication** is still required.

In other words:

- **IPsec AH / ESP** uses Network-layer security (IPv6 extension headers).
- **Authentication Trailer (RFC 7166)** uses OSPFv3 protocol-layer security.

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Both approaches protect OSPFv3 routing exchanges, but they do so at different layers and with different tradeoffs.

How Authentication Trailer Works at a High Level

When Authentication Trailer is enabled:

1. **OSPFv3 appends** an **authentication trailer** to each packet
2. A cryptographic hash (**HMAC**) is **calculated** over the packet contents using a process like OSPFv2 MD5 authentication, but with modern SHA-based algorithms.
3. The **AT-bit** (Authentication Trailer bit) is set in **OSPFv3 Hello** and **DBD** packets
4. **Neighbors** that **do not support or match** the authentication parameters will **not form adjacencies**

The result is cryptographically protected OSPFv3 routing updates without IPsec.

Configuration Model

Unlike IPsec-based authentication, the Authentication Trailer uses **Cisco IOS key chains**, similar to other routing protocols:

- A key chain defines the following:
  - The **hash algorithm** (for example, **HMAC-SHA-256**)
  - **One or more keys**
- The key chain is applied in the following ways:
  - Per interface

    or
  - Per OSPFv3 area

This makes Authentication Trailer familiar to administrators who have configured authentication for OSPFv2, EIGRP, or IS-IS.

Step 1: Remove OSPFv3 IPsec ESP Configurations on D1 and D2.

On both D1 and D2, remove the IPsec ESP configurations. If you do this fast enough, neighbor adjacencies should not be impacted.

```
D1(config)# interface GigabitEthernet1/0/1
D1(config-if)# no ospfv3 encryption ipsec spi 257 esp aes-cbc 128
12345678901234567890123456789012 sha1
12345678901234567890123456789012345678901234567890
*Jan 19 20:39:06.545: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF

D2(config)# interface GigabitEthernet1/0/1
D2(config-if)# no ospfv3 encryption ipsec spi 257 esp aes-cbc 128
12345678901234567890123456789012 sha1
12345678901234567890123456789012345678901234567890
*Jan 19 19:21:18.573: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
```

Step 2: Configure the Key Chain on D1 and D2.

A **key chain** is a structured way to define and manage cryptographic keys used for routing protocol authentication. Instead of configuring a single static password directly on an interface, key chains allow you to define one or more keys, each identified by a **key ID**, along with the cryptographic algorithm used to generate authentication information.

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Key chains are commonly used by routing protocols such as **OSPF, EIGRP, and IS-IS** to provide **cryptographic authentication and integrity** for routing updates.

a. Configure the key chain for D1.

```
D1(config)# key chain OSPF_KEYS
D1(config-keychain)# key 1
D1(config-keychain-key)# cryptographic-algorithm hmac-sha-256
D1(config-keychain-key)# key-string CRYPTOKEY
```

b. Configure the same key chain for D2.

```
D2(config)# key chain OSPF_KEYS
D2(config-keychain)# key 1
D2(config-keychain-key)# cryptographic-algorithm hmac-sha-256
D2(config-keychain-key)# key-string CRYPTOKEY
```

Explanation

- The **key chain name** (**OSPF_KEYS**) is a logical container for one or more authentication keys.
- **Key ID 1** uniquely identifies the key within the key chain.
- **HMAC-SHA-256** is used to generate a secure hash that provides authentication and integrity for OSPFv3 packets.
- The **key-string** (**CRYPTOKEY**) is the shared secret used to compute the HMAC and must be identical on both neighbors.
- **No send or accept lifetimes** are configured, so the key is **always valid by default**. When a key chain contains multiple keys, OSPF selects the key to use based on key lifetimes. The key with a valid send lifetime is used to authenticate outgoing packets, while any key with a valid accept lifetime may be used to verify incoming packets. This allows for overlapping keys during key rollover so that OSPF adjacencies are not disrupted. If no lifetimes are configured, all keys are always valid and the key with the highest key ID is used for sending.

Note: A key chain is configured once and can be applied to multiple OSPFv3 interfaces or areas, allowing the same authentication parameters to be reused without redefining them.

Why Use Key Chains?

Key chains provide several important advantages:

- **Stronger security** - They support modern cryptographic algorithms such as HMAC-SHA-256, which are significantly more secure than legacy methods like MD5.
- **Key rollover without downtime** - Multiple keys can be defined in a single key chain, allowing administrators to change keys without breaking routing adjacencies.
- **Clear separation of policy and application** - The key chain defines how authentication is performed, while OSPFv3 simply references the key chain where needed.
- **Consistency across interfaces and protocols** - The same key chain can be reused across multiple interfaces or applied to an entire OSPFv3 area.

Step 3: Apply the Key Chain to OSPFv3 Area 0

In this step, you will **enable OSPFv3 Authentication Trailer (RFC 7166)** by associating the previously configured key chain with **Area 0**. Applying the key chain at the **area level** ensures that cryptographic authentication is enabled for **all OSPFv3 interfaces in Area 0** and for **both IPv4 and IPv6 address families**.

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To clearly illustrate how OSPF authentication behaves, the key chain is first applied on **D1 only**. This creates an authentication mismatch, causing the OSPFv3 adjacency to drop. Once the same configuration is applied on **D2**, the adjacency is re-established.

a. On **D1**, enable OSPFv3 cryptographic authentication for Area 0 using the OSPF_KEYS key chain:

```
D1(config)# router ospfv3 123
D1(config-router)# area 0 authentication key-chain OSPF_KEYS
*Jan 19 20:50:17.473: %OSPFv3-5-ADJCHG: Process 123, IPv6, Nbr 1.1.1.3 on
GigabitEthernet1/0/1 from FULL to DOWN, Neighbor Down: Dead timer expired
*Jan 19 20:50:18.090: %OSPFv3-5-ADJCHG: Process 123, IPv4, Nbr 1.1.1.3 on
GigabitEthernet1/0/1 from FULL to DOWN, Neighbor Down: Dead timer expired
```

After this command is entered, OSPFv3 packets sent by D1 include an **Authentication Trailer**. On D2, you may see errors indicating that unauthenticated or **invalid packets** are being received due to a **Bad Checksum**. Because D2 is not yet configured to use the same key chain, the **OSPFv3 adjacency will also fail** if the dead timer expires before you configure the adjacent router.

b. Apply the same key chain to Area 0 on **D2**.

```
*Jan 19 19:32:16.575: %OSPFv3-4-ERRRCV: OSPFv3-123-IPv4 Received invalid packet: Bad
Checksum from FE80::D1:1, GigabitEthernet1/0/1
*Jan 19 19:32:19.301: %OSPFv3-5-ADJCHG: Process 123, IPv4, Nbr 1.1.1.2 on
GigabitEthernet1/0/1 from FULL to DOWN, Neighbor Down: Dead timer expired
*Jan 19 19:32:21.029: %OSPFv3-5-ADJCHG: Process 123, IPv6, Nbr 1.1.1.2 on
GigabitEthernet1/0/1 from FULL to DOWN, Neighbor Down: Dead timer expired
D2(config)# router ospfv3 123
D2(config-router)# area 0 authentication key-chain OSPF_KEYS
*Jan 19 19:41:36.147: %OSPFv3-5-ADJCHG: Process 123, IPv6, Nbr 1.1.1.2 on
GigabitEthernet1/0/1 from LOADING to FULL, Loading Done
*Jan 19 19:41:36.202: %OSPFv3-5-ADJCHG: Process 123, IPv4, Nbr 1.1.1.2 on
GigabitEthernet1/0/1 from LOADING to FULL, Loading Done
```

Once this configuration matches on both routers, you should see messages indicating that OSPFv3 **adjacency is automatically re-established** for **both address families**.

Explanation

- The **area 0 authentication key-chain** command enables **OSPFv3 Authentication Trailer** for the entire area.
- The configured key chain is used to generate and verify **HMAC-SHA-256** authentication data.
- Because authentication is applied at the **area level**, it automatically applies to all **OSPFv3 interfaces in Area 0**.
- OSPFv3 adjacencies fail when authentication parameters do not match on both neighbors and recover once they are consistent.
- Since OSPFv3 is running with **address families**, authentication is enforced for **both IPv4 and IPv6 adjacencies**, even though the protocol operates over IPv6.

Applying a key chain to an OSPFv3 area enables protocol-level cryptographic authentication without IPsec. All routers in the area must be configured with matching authentication parameters for OSPFv3 neighbor adjacencies to form.

Applying an OSPFv3 Authentication Trailer Key Chain at the Interface Level

In addition to applying a key chain at the **OSPFv3 area level**, OSPFv3 Authentication Trailer can also be enabled **per interface**. Applying authentication at the interface level allows you to secure **only a specific link**, rather than all interfaces in an area.

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This approach is useful when:

- Only certain links require authentication.
- Different interfaces need different authentication policies.
- You want more granular control over OSPFv3 security.

If authentication is configured both at the area level and the interface level, the interface-level configuration takes precedence.

The following is an example of how to apply the OSPF_KEYS key chain directly to an OSPFv3-enabled interface:

```
Router(config)# interface GigabitEthernet1/0/1
Router(config-if)# ospfv3 authentication key-chain OSPF_KEYS
```

Step 4: Verify neighbor adjacency and reachability after OSPFv3 AH Trailer Authentication

a. Verify D1's neighbor adjacency was re-established with D2.

```
D1# show ospfv3 neighbor

          OSPFv3 123 address-family ipv4 (router-id 1.1.1.2)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
1.1.1.3           0   FULL/  -        00:00:32    9               GigabitEthernet1/0/1

          OSPFv3 123 address-family ipv6 (router-id 1.1.1.2)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
1.1.1.3           0   FULL/  -        00:00:30    9               GigabitEthernet1/0/1
```

b. Verify D1's reachability to the remote IPv4 network on D2.

```
D1# show ip route ospfv3 | begin Gateway
Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 7 subnets, 3 masks
O        172.16.10.0/24 [110/2] via 172.16.1.2, 06:08:35, GigabitEthernet1/0/1
D1#
```

c. Verify D1's reachability to the remote IPv6 network on D2.

```
D1# show ipv6 route ospf | begin 2001
O   2001:DB8:ACAD:A010::/64 [110/2]
     via FE80::D2:1, GigabitEthernet1/0/1
```

Step 5: Verify the OSPFv3 Authentication Trailer key chain.

Verify that the **key chain** used for OSPFv3 Authentication Trailer has been configured correctly using the **show key chain** command. The **show key chain** command displays all key chains defined on the router, including key IDs, key strings, and lifetime information.

```
D1# show key chain
Key-chain OSPF_KEYS:
    key 1 -- text "CRYPTOKEY"
        cryptographic-algorithm: hmac-sha-256
        accept lifetime (always valid) - (always valid) [valid now]
```

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```
        send lifetime (always valid) - (always valid) [valid now]
```

Explanation

- **Key-chain OSPF_KEYS** confirms that the key chain **OSPF_KEYS** has been created successfully and is used by OSPFv3 for cryptographic authentication.
- **Key 1 -- text "CRYPTOKEY"** is the only key defined in the key chain and will be used for authentication. It defines **key ID 1** with the shared secret **CRYPTOKEY** that neighboring routers must match to authenticate OSPFv3 packets.
- **cryptographic-algorithm: hmac-sha-256** specifies that **HMAC** with the **SHA-256 hashing algorithm** is used to authenticate and protect the integrity of OSPFv3 packets.
- The **accept lifetime** and **send lifetime** are shown as **always valid**, indicating that no explicit lifetimes were configured This indicates that this key is always accepted for verifying received and sent OSPFv3 packets.
- Because the key is currently valid for both sending and receiving, it can be used immediately by OSPFv3 Authentication Trailer.

This output verifies that the key chain is properly configured and ready to be applied to OSPFv3 interfaces or areas.

Step 6: Verify OSPFv3 AH Authentication Trailer on the interface.

In this step, you will verify that **OSPFv3 Authentication Trailer (RFC 7166)** is enabled and functioning on the OSPFv3 interface. The show ospfv3 interface command confirms that cryptographic authentication is active, identifies the key chain and algorithm in use, and shows that neighbor adjacencies are successfully formed.

On **D1**, enter the following command:

```
D1# show ospfv3 interface GigabitEthernet1/0/1
GigabitEthernet1/0/1 is up, line protocol is up (connected)
  Link Local Address FE80::D1:1, Interface ID 9 (snmp-if-index)
  Internet Address 172.16.1.1/30
  Area 0, Process ID 123, Instance ID 64, Router ID 1.1.1.2
  Network Type POINT_TO_POINT, Cost: 1
  Cryptographic authentication enabled
    Sending SA: Key 1, Algorithm HMAC-SHA-256 - key chain OSPF_KEYS

  <output omitted>

  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.3
  Suppress hello for 0 neighbor(s)
GigabitEthernet1/0/1 is up, line protocol is up (connected)
  Link Local Address FE80::D1:1, Interface ID 9 (snmp-if-index)
  Area 0, Process ID 123, Instance ID 0, Router ID 1.1.1.2
  Network Type POINT_TO_POINT, Cost: 1
  Cryptographic authentication enabled
    Sending SA: Key 1, Algorithm HMAC-SHA-256 - key chain OSPF_KEYS

  <output omitted>

  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.3
```

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```
  Suppress hello for 0 neighbor(s)
D1#
```

Examining the Output

Several important sections of the output confirm that Authentication Trailer–based cryptographic authentication is correctly configured:

- **Cryptographic authentication enabled**: This line confirms that OSPFv3 Authentication Trailer is active on the interface. Unlike IPsec AH or ESP, this authentication is performed **within the OSPFv3 protocol itself**, not at the IPv6 layer.
- **Sending SA: Key 1, Algorithm HMAC-SHA-256 - key chain OSPF_KEYS**: This line shows:
  - The **key ID (1)** currently being used
  - The **HMAC-SHA-256** algorithm selected for authentication
  - The **key chain (OSPF_KEYS)** supplying the cryptographic parameters

  Together, these confirm that the correct key chain and algorithm are in use.
- **Adjacent with neighbor** - The presence of an adjacent neighbor confirms that both routers are using matching authentication parameters. If the key chain, algorithm, or key value did not match, the adjacency would fail.
- **Output appears twice (address families)** - As in previous OSPFv3 verification steps, the output is shown twice - Once for the IPv4 address family and once for the IPv6 address family. This confirms that **Authentication Trailer is enforced for both address families**, even though OSPFv3 continues to operate over IPv6.

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